Finding the Edge Cases That Crash Self-Driving Cars

Failure discovery read from inside a vision-language-action driving model, built to work with any VLA

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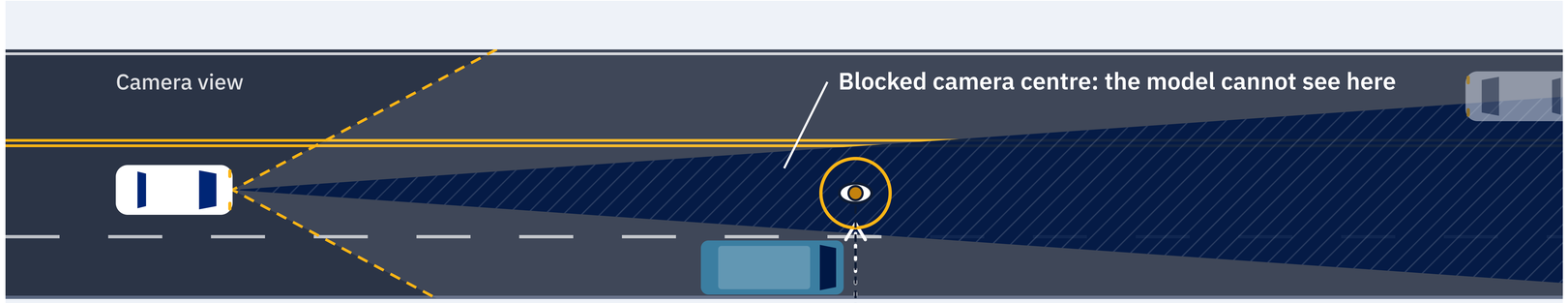
Test lists only catch the crashes someone imagined

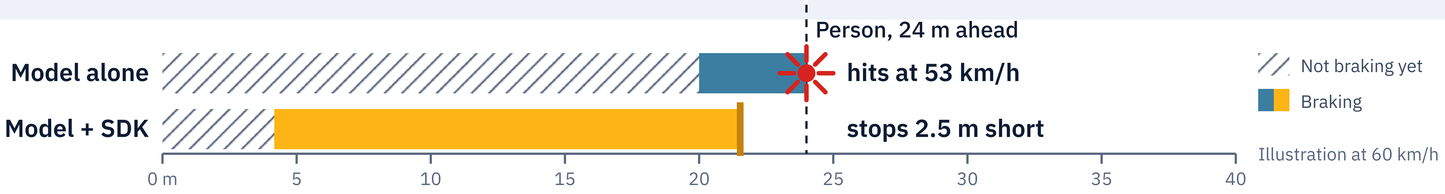
Driving models such as SimLingo are checked against failures people already thought of. We find new ones by reading the model's own internal signals, with no list. To prove it works we plant a failure on purpose: block the centre of the camera.

3.6×

more failures with the camera centre blocked

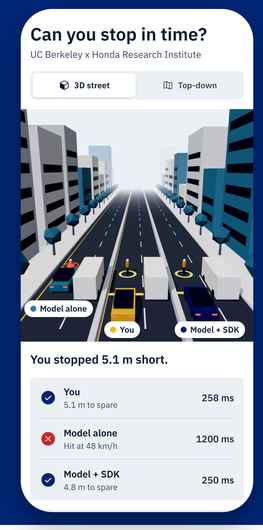
43 of 60 routes against 12 of 60 clear, in CARLA





Can you stop in time?

Scan and drive: a person steps out from behind a van, and you hold Brake. Then see the same moment for the model alone and for the model trained on what we found.



failure-axis-status.vercel.app/drive

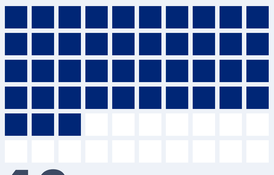
How discovery works



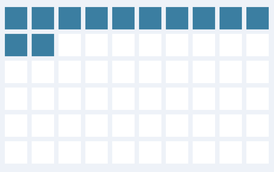
We record layer 10 of SimLingo on every drive and contrast failing drives with clean ones. Each direction that separates them is a failure axis, and it names the scenarios to train on next.

Planted failures must really fail

43 of 60
Centre blocked



12 of 60
Clear camera



31 routes failed only when blocked, 0 only when clear ($p = 2 \times 10^{-8}$). Fog and animals look unusual but did not raise failures, so they are decoys.

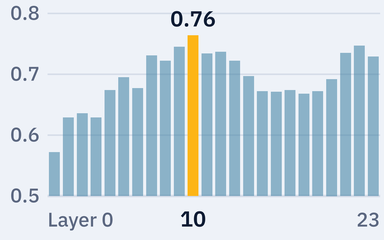


Dense fog: 15 of 59 failed, against 18 of 60



Animals crossing: 1 of 9 failed, against 2 of 10

Where to listen



We scored all 24 layers on 212 drives for how well each flags the 61 failures. Layer 10 is best, where 0.5 is chance.

Control check



51 planted of 73 failures

Every method finds this easy planted failure: the bench works, but ours has not won yet. Next: harder failures and a second VLA.

SimLingo: Renz et al., CVPR 2025. Bench2Drive: Jia et al., NeurIPS 2024. CARLA: Dosovitskiy et al., CoRL 2017. Overlap is Jaccard similarity, shared failures over combined failures.

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